

CV

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Ayal Klein

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Birth: 10.8.1988, Illinois, U.S.A

Education

2025-Currently: Senior Faculty (Lecturer), Department of Computer Science, Ariel University.

2024-2025: Postdoc at Queen Mary University, London. Studying how can AI be leveraged to improve clinical psychology and mental healthcare.

2019-2024: Ph.D. in Computer Science at Bar Ilan University.

Thesis title: Representing Natural Language Information with Natural Language Questions and Answer. at BIU Natural Language Processing Lab, advised by Prof. Ido Dagan.

2016-2018: M.Sc. in Computer Science at Bar Ilan University. Advised by Prof. Ido Dagan.

2013-2015: B.A Studies in Psychology (major) and Philosophy (secondary).

2007-2012: Yeshivat Hesder at Petah-Tikva.

2002-2006: HaDarom High-School-Yeshiva, Rehovot.

Awards and Honors

2015: B.A Graduation with Great Honors. Average Grade: 95.9

2013-2015: Student at Bar Ilan's Program for Honor Scholars.

2013: Dean's List (Faculty of Social Sciences).

Activities

- Program Committee for top-tier ACL venues – EMNLP, ACL, NAACL, EACL & ACL ARR track (2020-current)
- Internship at Amazon (2022), Alexa Shopping Research, working on computational humor generation using large language models.
- Internship at Intel Labs (2020), working on semantically-enhanced domain transfer for aspect-based sentiment analysis.
- Check Point Software Technologies Ltd. Part-time Software Developer (2015-2017).
- Research Assistant at Language and Creativity Lab, Neuro-Science Department, Bar Ilan University (2014-2015).

Educational Work

- Lecturer, "Object Oriented Programming" Course, School of Computer Science, Ariel University (2026)

- Lecturer, "Advanced Models of Language Understanding" Course, School of Computer Science, Ariel University (2025)
- Lecturer, "Databases" Course, School of Computer Science, Ariel University (2025)
- Lecturer, "Artificial Neural Networks" Course, Neuro-Science department, Bar Ilan University (2023-2024)
- Teacher Assistant for "Object Oriented Programming" Course (89-111), Bar Ilan University (2019-2020).
- Teacher Assistant for "Discrete Mathematics 1" Course (89-195), Bar Ilan University (2019).
- Teacher Assistant for "Text Understanding" Course (89-565), Bar Ilan University (2018).

Interests, Skills, Courses

- Main areas of interest: Natural Language Processing, Deep Learning, Interpretability, Data Science, Computational Social Science, NLP for Health.
- Programming: Python, C, C++, Java, Scala
- Languages: Hebrew Native Speaker. Excellent English.
- Soft Skills: Creative and analytical thinking. Excellent expression ability.

Presentations

- Plenary talk, "LLMs for Clinical Insight: Analyzing Model-Generated Clinical Summaries of Social Media Timelines" (IDSAI 2024)
- Seminar Lecture, "Short overview of modern NLP models", Advanced Seminar in Data Science and Neuroscience (Neuroscience, Bar-Ilan University, 2023, 2024)
- Plenary talk, "QASem Parsing: Text-to-text Modeling of QA-based Semantics" (BISFAI 2023)
- Plenary talk, "QA-Align: Representing Cross-Text Content Overlap by Aligning Question-Answer Propositions" (ISCOL 2021).
- Pitch talk, "QASem - NL-based Semantic Representation" (Dagstuhl Seminar, Multi-Document Information Consolidation, Germany 2019)

Publications

17. Bar-Shachar, Yael, Dana Rafael, Anmol Goel, Ayal Klein, Iryna Gurevych, and Dana Atzil-Slonim. "Evaluating Large Language Models versus Human Therapists in Generating Interventions for Patients with Depression in Psychodynamic Psychotherapy." *Journal of Consulting and Clinical Psychology*, in press. Accepted August 17, 2026.
16. Keren, Liran, Yael Bar-Shachar, Ayal Klein, Maria Liakata, and Dana Atzil-Slonim. "A Comparative Evaluation of LLM-Generated and Clinician-Written Psychotherapy Session Summaries." *Psychotherapy Research*, in press. 2026.
15. Ali, Iqra, Talia Tseriotou, Guy Dvir, Callum Chan, Yuxiang Zhou, Juan Antonio Lossio-Ventura, Ayal Klein, et al. "Overview of the CLPsych 2026 Shared Task: Capturing and Characterizing Mental Health Changes through Social Media Timeline Dynamics." In *Proceedings of the 10th Workshop on Computational Linguistics and Clinical Psychology (CLPsych 2026)*, pp. 389–421. 2026.

14. Davidov, Jonathan, Aviv Slobodkin, Shmuel Tomi Klein, Reut Tsarfaty, Ido Dagan, and Ayal Klein. "Effective QA-driven Annotation of Predicate-Argument Relations Across Languages." Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026, Long Papers), pp. 2484–2502, Rabat, Morocco. 2026.
13. Klein, Ayal, Jiayu Song, Jenny Chim, Liran Keren, Andreas Triantafyllopoulos, Björn W. Schuller, Maria Liakata, and Dana Atzil-Slonim. "Clinical Summaries of Social Media Timelines for Mental Health Monitoring: Human vs. Large Language Model Comparative Evaluation." JMIR Formative Research 2026; 10:e71230. doi:10.2196/71230. 2026.
12. Tseytlin, Maria, Paul Roit, Omri Abend, Ido Dagan, and Ayal Klein. "QA-Noun: Representing Nominal Semantics via Natural Language Question-Answer Pairs." Proceedings of the International Joint Conference on Natural Language Processing and the Asian Conference on Natural Language Processing (IJCNLP-AACL 2025), pp. 2727–2741, Mumbai, India. 2025.
11. Tseriotou, Talia, Jenny Chim, Ayal Klein, Aya Shamir, Guy Dvir, Iqra Ali, Cian Kennedy, Guneet Singh Kohli, Anthony Hills, Ayah Zirikly, Dana Atzil-Slonim, and Maria Liakata. "Overview of the CLPsych 2025 Shared Task: Capturing Mental Health Dynamics from Social Media Timelines." In Proceedings of the 10th Workshop on Computational Linguistics and Clinical Psychology (CLPsych 2025), pages 193–217, Albuquerque, New Mexico. Association for Computational Linguistics. 2025.
10. Zhang, Shiyue, David Wan, Arie Cattán, Ayal Klein, Ido Dagan, and Mohit Bansal. "QAPyramid: Fine-grained Evaluation of Content Selection for Text Summarization." In Proceedings of the Conference on Language Modeling (COLM 2025). Association for Computational Linguistics. 2025.
9. Triantafyllopoulos, Andreas, Yannik Terhorst, Iosif Tsangko, Florian B. Pokorny, Ayal Klein, et al. "Large language models for mental health." arXiv preprint arXiv:2411.11880. 2024.
8. Roit, Paul, Aviv Slobodkin, Eran Hirsch, Arie Cattán, Ayal Klein, Valentina Pyatkin, and Ido Dagan. "Explicating the Implicit: Argument Detection Beyond Sentence Boundaries." Proceedings of ACL 2024 (Volume 1: Long Papers), pp. 16394–16409, Bangkok, Thailand. 2024.
7. Pesahov, Leon, Ayal Klein, and Ido Dagan. "QA-Adj: Adding Adjectives to QA-based Semantics." In Proceedings of the Fourth International Workshop on Designing Meaning Representations, pp. 74–88. 2023.
6. Klein, Ayal, Eran Hirsch, Ron Eliav, Valentina Pyatkin, Avi Caciularu, and Ido Dagan. "QASem Parsing: Text-to-text Modeling of QA-based Semantics." In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, pp. 7742–7756. 2022.
5. Klein, Ayal, Oren Pereg, Daniel Korat, Vasudev Lal, Moshe Wasserblat, and Ido Dagan. "Opinion-based Relational Pivoting for Cross-domain Aspect Term Extraction." In Proceedings of the 12th Workshop on Computational Approaches to Subjectivity, Sentiment & Social Media Analysis, pp. 104–112. 2022.
4. Weiss, Daniela Brook, Paul Roit, Ayal Klein, Ori Ernst, and Ido Dagan. "QA-Align: Representing Cross-Text Content Overlap by Aligning Question-Answer Propositions." In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, pp. 9879–9894. 2021.

3. Klein, Ayal, Jonathan Mamou, Valentina Pyatkin, Daniela Stepanov, Hangfeng He, Dan Roth, Luke Zettlemoyer, and Ido Dagan. "QANom: Question-Answer driven SRL for Nominalizations." In Proceedings of the 28th International Conference on Computational Linguistics, pp. 3069-3083. 2020.
2. Roit, Paul, Ayal Klein, Daniela Stepanov, Jonathan Mamou, Julian Michael, Gabriel Stanovsky, Luke Zettlemoyer, and Ido Dagan. "Controlled Crowdsourcing for High-Quality QA-SRL Annotation." In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 7008-7013. 2020.
1. Pyatkin, Valentina, Ayal Klein, Reut Tsarfaty, and Ido Dagan. "QADiscourse—Discourse Relations as QA Pairs: Representation, Crowdsourcing and Baselines." In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 2804-2819. 2020.